

1. Rewrite A as $\begin{bmatrix} a & 1 & b \\ 2 & 3 & 4 \\ c & 5 & d \end{bmatrix}$. Then $1 = C_{11} = \begin{vmatrix} 3 & 4 \\ 5 & d \end{vmatrix} = 3d - 20$ and hence $d = 7$.

Now $2 = C_{12} = -\begin{vmatrix} 2 & 4 \\ c & d \end{vmatrix} = -(2d - 4c) = -14 + 4c$ and hence $c = 4$. So

$$C_{13} = \begin{vmatrix} 2 & 3 \\ c & 5 \end{vmatrix} = (2)(5) - 3c = 10 - 12 = -2.$$

2. (a) Note that $A(\text{adj } A) = (\det A)I$. Taking determinant on both sides, this gives

$$(\det A)(\det(\text{adj } A)) = (\det A)^n.$$

Since A is invertible, $\det A \neq 0$. So we have

$$\det(\text{adj } A) = (\det A)^{n-1}.$$

- (b) Note that

$$A(\text{adj } A) = (\det A)I.$$

The same formula also holds if we replace A by $\text{adj } A$. Therefore, we have

$$(\text{adj } A)(\text{adj}(\text{adj } A)) = (\det(\text{adj } A))I = (\det A)^{n-1}I$$

by part (a). In view of these two equations,

$$\text{adj}(\text{adj } A) = (\det A)^{n-1}(\text{adj } A)^{-1} = (\det A)^{n-1} \cdot \frac{A}{\det A} = (\det A)^{n-2}A.$$

3. (a) The inverse of M is

$$P = \begin{bmatrix} A^{-1} & 0_{n \times m} \\ -B^{-1}CA^{-1} & B^{-1} \end{bmatrix}$$

because using block multiplication,

$$PM = \begin{bmatrix} A^{-1}A + 0_{n \times m}C & A^{-1}0_{n \times m} + 0_{n \times m}B \\ (-B^{-1}CA^{-1})A + B^{-1}C & (-B^{-1}CA^{-1})0_{n \times m} + B^{-1}B \end{bmatrix} = \begin{bmatrix} I_n & 0_{n \times m} \\ 0_{m \times n} & I_m \end{bmatrix}.$$

- (b) We set $A = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$ in (a). Then $A^{-1} = \begin{pmatrix} -4 & 3 \\ 3 & -2 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} -1 & 1 \\ 3 & -2 \end{pmatrix}$. From (a), the inverse of the matrix is

$$\begin{pmatrix} -4 & 3 & 0 & 0 \\ 3 & -2 & 0 & 0 \\ 4 & -3 & -1 & 1 \\ -10 & 7 & 3 & -2 \end{pmatrix}.$$

4. Note that $(A^T)^2 = I_5$ implies $A^2 = I_5$ and $\det A = \det A^T = \pm 1$.

(a) Proof: $\det(A^T A) = (\det A)^2 = 1$.

(b) Disproof: Take $A = I_5$, then $(A^T)^2 = I_5$ and $\det A^2 A^T = 1 \neq -1$.

(c) Proof: As $A \operatorname{adj} A = (\det A)I_5$, we have

$$\det(A^2 \operatorname{adj} A) = \det A \cdot \det[(\det A)I_5] = (\det A)^6 = 1.$$

(d) Disproof: Take $A = -I_5$, then $(A^T)^2 = I_5$ and $\operatorname{adj} A = I_5 \neq A$.

5. (a) Disproof by counter example: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is elementary, but $2\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}$ is not symmetric.

(b) Disproof by counter example: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ are elementary, but

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \text{ is not.}$$